

Does Education Raise Adult Income in China?

Evidence from CFPS 2022

Empirical Essay

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Abstract

This paper examines the association between education and income among working-age adults in China using the CFPS 2022 analytic sample. The sample is restricted to adults aged 18–64 with non-missing wage income and core covariates, yielding 4,699 observations. The main dependent variable is $\ln(\text{wage} + 1)$, and the key explanatory variable is years of education. Ordinary least squares models with robust standard errors are estimated sequentially with demographic controls and province fixed effects. The preferred specification estimates an education coefficient of 0.0882, significant at the 1 percent level, implying that one additional year of schooling is associated with roughly 8.8 percent higher wage income, conditional on observed controls. Robustness checks using education categories, an alternative household-income outcome, and a positive-wage subsample preserve the positive association. Because the design relies on cross-sectional observational data, the estimates should be interpreted as conditional associations rather than strict causal effects.

Keywords: education, income, CFPS, China, Mincer equation, returns to schooling

1 Introduction

Education is central to modern labor-market inequality because it can raise productive skills, improve job matching, and expand access to higher-paying occupations. The human-capital tradition treats schooling as an investment that can increase future earnings [Becker, 1964, Mincer, 1974]. At the same time, estimating the effect of education is empirically difficult because schooling choices are correlated with family background, ability, location, and labor-market opportunities [Card, 1999, Angrist and Krueger, 1991].

This paper asks a focused empirical question: in the CFPS 2022 adult sample, is higher education associated with higher income after controlling for observable demographic and regional differences? The answer is yes. The preferred model shows a large and statistically precise positive association between years of education and log wage income. The analysis, however, is intentionally cautious about interpretation: the results document a robust conditional correlation, not a causal return to schooling.

2 Literature

The empirical specification follows the Mincerian earnings-function framework, in which log earnings are modeled as a function of schooling and labor-market experience [Mincer, 1974]. Human-capital theory provides the economic mechanism behind this relationship: schooling may raise earnings by increasing skills and productivity [Becker, 1964]. Later work emphasizes that OLS estimates may be biased if unobserved ability or family background affects both education and earnings; quasi-experimental approaches are therefore important for causal interpretation [Card, 1999, Angrist and Krueger, 1991, Heckman et al., 2008].

Evidence from China also finds substantial education-income gradients, especially during market-oriented reforms. Zhang et al. [2005] document rising economic returns to schooling in urban China from 1988 to 2001. This paper contributes a compact update using CFPS 2022 microdata, a nationally oriented survey whose sampling design is documented by Xie and Lu [2015].

3 Data and Variables

The analysis uses CFPS 2022 data from the Institute of Social Science Survey at Peking University [Institute of Social Science Survey, Peking University, 2022]. The sample keeps 2022 respondents aged 18–64, requires non-missing wage income, and drops observations missing the main outcome, education, demographic controls, medical-insurance indicator, or province code. The final analytic sample contains 4,699 observations.

The dependent variable is $\ln(\text{wage} + 1)$, where wage is annual individual wage income. The core explanatory variable is years of education. Controls include age, age squared, gender, marital status, rural hukou, medical insurance, and province fixed effects. Table 1 reports descriptive statistics, and Table 2 reports the education composition of the analytic sample.

Table 1: Descriptive Statistics

Variable	N	Mean	SD	Min	Median	Max
Annual wage income	4,699	60594.132	68985.459	0.000	48000.000	1800000.000
$\ln(\text{wage} + 1)$	4,699	10.489	1.567	0.000	10.779	14.403
Years of education	4,699	11.006	4.376	0.000	12.000	22.000
Age	4,699	40.867	10.921	18.000	40.000	64.000
Male	4,699	0.575	0.494	0.000	1.000	1.000
Rural hukou	4,699	0.795	0.404	0.000	1.000	1.000

Table 2: Education Distribution

Education category	N	Percent
Illiterate	243	5.17
Primary	589	12.53
Junior high	1,463	31.13
High school	774	16.47
College	702	14.94
Bachelor	824	17.54
Master	91	1.94
Doctorate	13	0.28

4 Empirical Strategy

The baseline model is

$$\ln(\text{wage}_i + 1) = \alpha + \beta \text{educ}_i + \gamma X_i + \mu_p + \varepsilon_i, \quad (1)$$

where educ_i is years of education, X_i contains individual controls, and μ_p denotes province fixed effects. Robust standard errors are used throughout. Three baseline specifications are estimated: a bivariate model, a model adding demographic controls, and a preferred model adding province fixed effects. Robustness checks replace the income outcome with $\ln(\text{inc1} + 1)$ and restrict the sample to respondents with positive wage income.

5 Results

Figure 1 shows that junior-high, high-school, college, and bachelor's groups account for most of the analytic sample. Figure 2 displays a clear upward gradient in mean log wage income across education categories. The same pattern appears when education is represented as years of schooling in Figure 3.

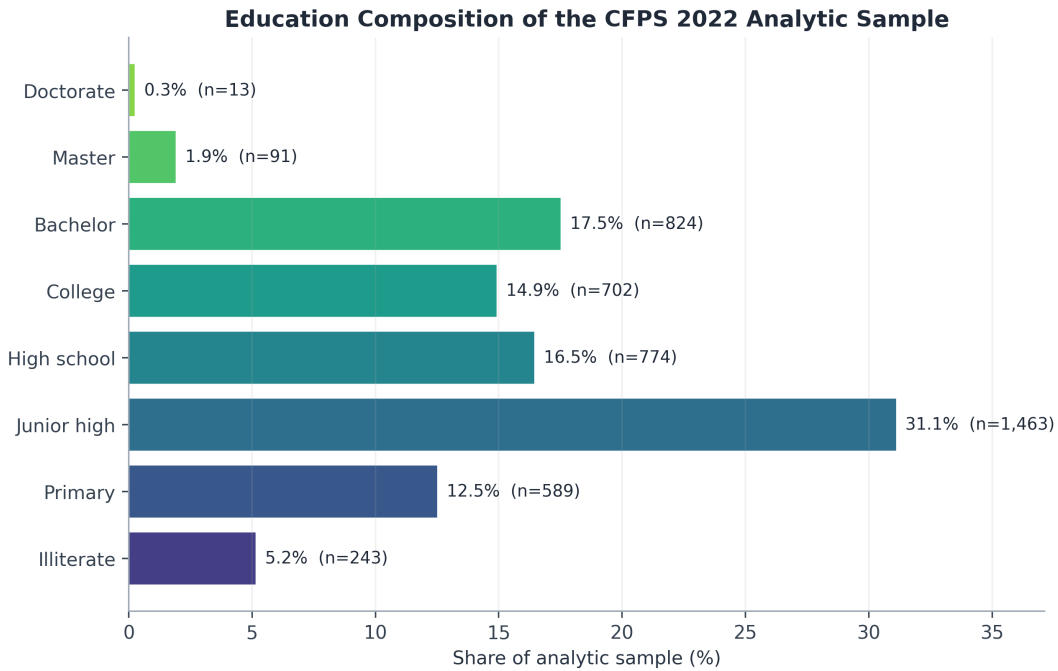


Figure 1: Education composition of the analytic sample

Figure 1 shows that respondents are mainly distributed across junior high school, senior high school, college, and bachelor's degree categories. Notably, respondents with college education or above account for roughly one-third of the analytic sample, reflecting the broader accessibility of higher education in recent years.

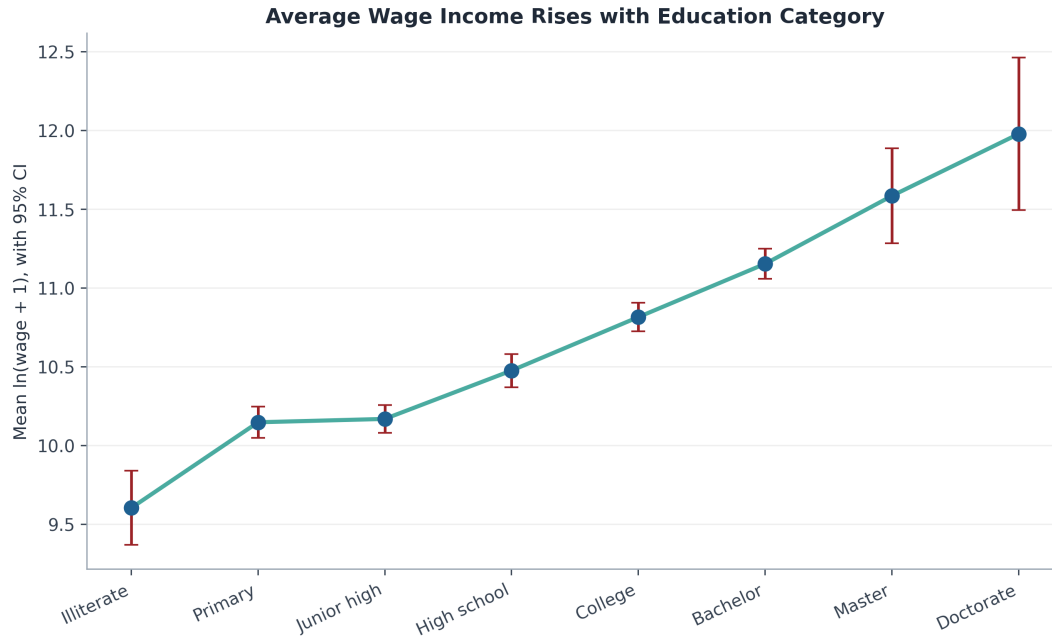


Figure 2: Mean log wage income by education category

Figure 2 further illustrates that average log wage income rises with educational attainment. The increase is especially visible from junior high school to senior high school and from senior high school to college, while a positive wage premium is also observed from college to bachelor's degree. This pattern provides descriptive evidence for a positive education-income association.

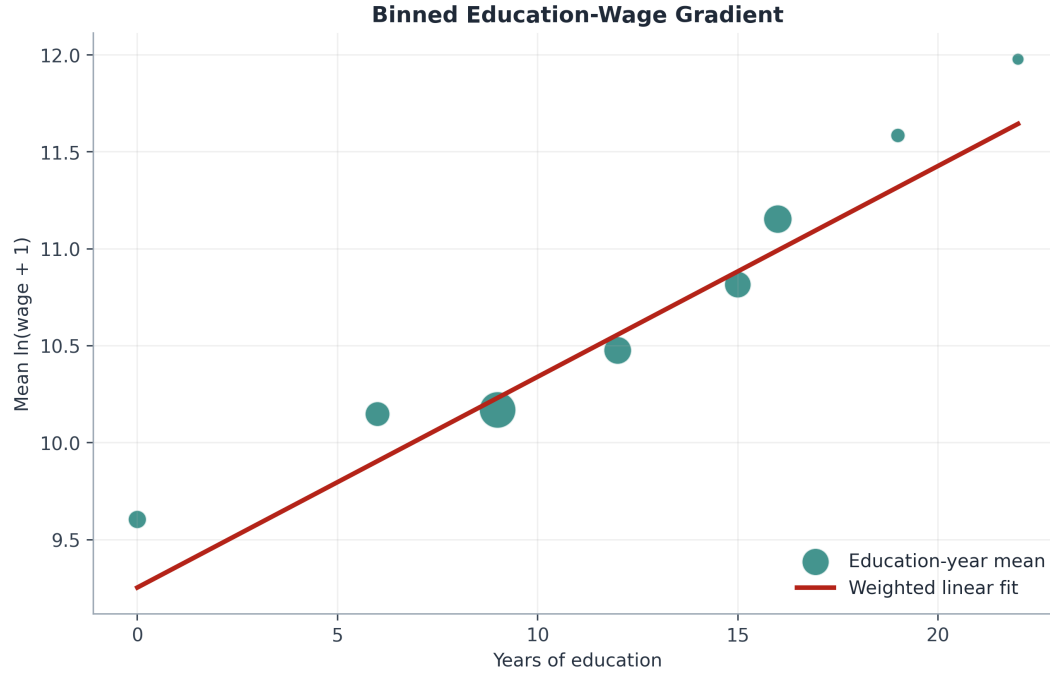


Figure 3: Binned education-wage gradient

Figure 3 presents a similar positive gradient based on grouped mean values by years of schooling. This graphical evidence is consistent with Figure 2 and reinforces the same relationship using education as a continuous variable.

Table 3 presents the main regression estimates. The preferred specification with demographic controls and province fixed effects estimates an education coefficient of 0.0882 with a robust standard error of 0.0062. The coefficient is significant at the 1 percent level. Interpreted approximately, one additional year of education is associated with about 8.8 percent higher wage income, conditional on the included controls.

Table 3: Education Coefficients Across Specifications

Model	Education coefficient	Robust SE	N	R^2
M1	0.0996***	(0.0052)	4,699	0.077
M2	0.0959***	(0.0063)	4,699	0.127
M3	0.0882***	(0.0062)	4,699	0.158
Alt. income	0.0728***	(0.0030)	4,699	0.340
Positive wage	0.0788***	(0.0037)	4,638	0.297

Notes: Robust standard errors are in parentheses. M1 includes only years of education. M2 adds demographic controls. M3 adds province fixed effects. Alt. income uses $\ln(\text{inc1} + 1)$ as the dependent variable. Positive wage restricts the sample to $\text{wage} > 0$. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

5.1 Hypothesis Testing

To formally test the relationship between education and income, this paper conducts a standard hypothesis test on the coefficient of years of schooling in the preferred model. The null hypothesis is $H_0 : \beta_{\text{educ}} = 0$, meaning that years of schooling have no statistically significant association with income after controlling for the other variables. The alternative hypothesis is $H_1 : \beta_{\text{educ}} \neq 0$.

Based on Model 3, the estimated coefficient is $\hat{\beta}_{\text{educ}} = 0.0882$ and the robust standard error is 0.0062. The implied test statistic is approximately

$$t = \frac{0.0882}{0.0062} \approx 14.23,$$

with $p < 0.001$. At the 1 percent significance level, the null hypothesis is rejected. The sample therefore provides strong statistical evidence that years of schooling are positively associated with income after controlling for age, gender, marital status, hukou status, medical insurance, and province fixed effects. As a comparison, applying the same logic to marital status in Model 3 yields a p-value of about 0.749, so the relationship between marital status and income is not statistically significant in this sample.

Figure 4 summarizes the education coefficients and their 95 percent robust confidence intervals. The positive association remains stable after adding controls and remains positive in both robustness checks. The alternative household-income outcome produces a smaller coefficient of 0.0728, while the positive-wage subsample gives 0.0788.

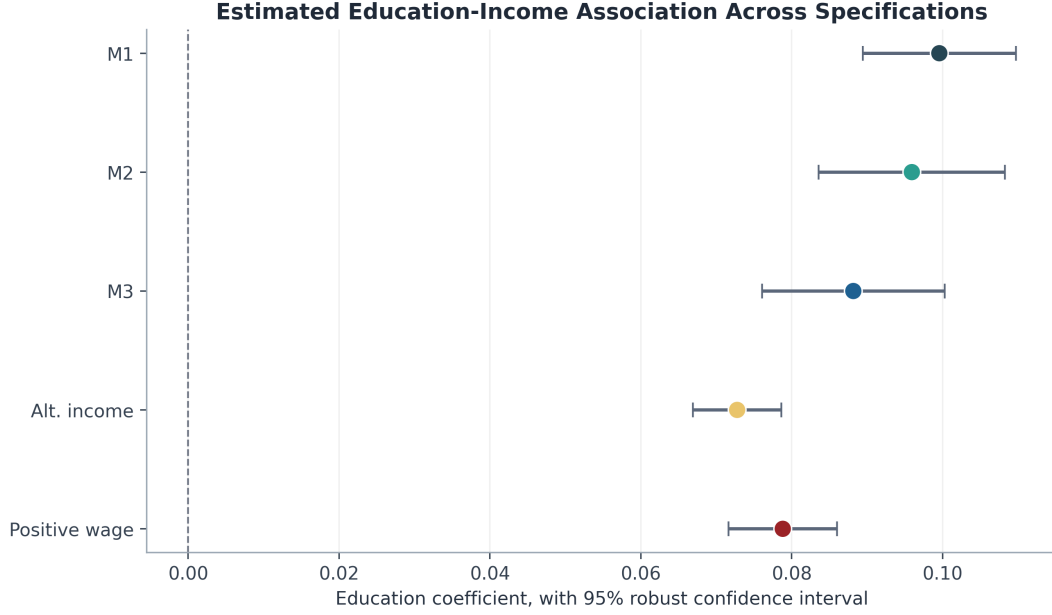


Figure 4: Education-income association across specifications

6 Discussion and Limitations

Based on CFPS 2022 data, this paper finds a significant and robust positive correlation between years of schooling and income among Chinese adults. After controlling for demographic variables and province fixed effects, each additional year of education is associated with an average wage increase of approximately 8.8 percent. This finding is consistent with human-capital theory and with recent empirical studies based on Chinese data. Several points deserve further discussion.

6.1 Correlation Versus Causation

The cross-sectional OLS method used in this paper cannot fully overcome omitted-variable bias from unobserved factors such as ability, motivation, family background, school quality, and occupational choice. Therefore, the 8.8 percent estimate should be understood as a conditional association rather than a strict rate of return to education. Although the robustness checks support the stability of the result's direction, future research aiming for causal inference could consider quasi-experimental designs such as instrumental variables based on compulsory-education reforms, regression-discontinuity designs, or difference-in-differences methods.

6.2 Potential Heterogeneity

This paper estimates an average association, but education returns may vary across groups. Important extensions include the following:

- **Gender:** Are returns to education higher for women than for men?
- **Hukou status:** Is there a gap between agricultural and non-agricultural hukou holders?
- **Region:** Do returns to education differ across eastern, central, and western China?
- **Education level:** Are marginal returns nonlinear between basic education and higher education?

Due to space constraints and the focus of this analysis, this paper does not conduct a systematic heterogeneity analysis, but this should be an important direction for future research.

6.3 Labor-Market Structure and Institutions

The income effect of education is not determined only by the supply side of human capital; it also depends on labor-market demand. If high-quality jobs are scarce, or if hukou segmentation, industry monopolies, or employment discrimination remain severe, the realized returns to education may be suppressed. Education policy should therefore be coordinated with industrial policy, employment policy, and hukou reform to more fully realize the income-enhancing potential of human capital.

6.4 Measurement and Representativeness

The wage income used in this paper is self-reported and may contain measurement error. In addition, the analytic sample is limited to adults with wage income and non-missing control variables, potentially excluding people without wage income, such as self-employed workers, farmers, and unemployed individuals. This limitation may affect the external validity of the conclusions. Future research could extend the analysis to broader income measures, such as total personal income or household per-capita income, and to more diverse employment types.

6.5 Contribution and Reproducibility

Despite these limitations, the value of this paper lies in its use of relatively recent CFPS 2022 microdata and a clearly structured, reproducible workflow that includes Stata, Python, and LaTeX files. By distinguishing correlation from causation, the paper provides a benchmark framework that can be directly extended or modified in subsequent research.

7 Policy Implications

The findings indicate that education remains a key factor affecting income levels in China’s labor market. Based on the estimated education-income association of approximately 8.8 percent, three policy implications follow.

First, continued investment in education has clear economic rationale. For individuals, education remains an effective path to income growth. For public policy, sustaining universal basic education and steadily expanding access to senior high school and higher education can improve the overall quality of the labor force and create conditions for narrowing income gaps.

Second, education policy should be coordinated with labor-market policy. Education can raise income only when the labor market efficiently allocates and rewards skilled workers. If industrial upgrading lags behind, high-quality jobs are insufficient, or institutional segmentation persists, the human-capital value of education may not be fully realized. Educational investment should therefore be aligned with employment promotion, industrial upgrading, and coordinated regional development.

Third, policymakers should pay attention to group differences in returns to education. This paper estimates an average relationship, but prior literature suggests that returns may vary by gender, hukou status, region, and education level. Future policy design should use more localized evidence, avoid one-size-fits-all interventions, and direct educational resources toward groups and sectors with greater return potential.

8 Conclusion

Using CFPS 2022 data, this paper finds a robust positive relationship between education and adult income in China. In the preferred model, one more year of education is associated with approximately 8.8 percent higher wage income after controlling for demographic factors and province fixed effects. The evidence supports the view that education is closely linked to labor-market advantage, but the cross-sectional design means the estimate should not be

read as a definitive causal return to schooling.

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